AI TRUST CASE STUDIES

Case Study 2: COMPAS Recidivism Algorithm

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GitHub Repository

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RESEARCH INTERESTS

Trustworthy AI, Human–AI Interaction (HAI), Responsible AI, Explainable AI (XAI), Human-Centered AI (HCAI), AI Ethics.

PURPOSE

This document is part of the AI Trust Case Studies project. It provides an educational analysis of a real-world AI system using widely accepted Trustworthy AI principles.

DISCLAIMER

This material is intended for educational and research purposes only. It is based on publicly available information from official reports, peer-reviewed literature, and reputable sources. The analysis reflects the cited evidence and does not represent the views of any organization mentioned. Readers should consult the original references for the most authoritative and up-to-date information.

CASE STUDY 2: COMPAS RECIDIVISM ALGORITHM

Topic: Fairness, Transparency, Accountability, and Trustworthy AI

LEARNING OBJECTIVES

After studying this case, you will be able to:

- Understand how AI is used in criminal justice decision-making.
- Explain the importance of fairness in AI-based risk assessment.
- Understand why transparency is essential for building trust in AI systems.
- Identify the role of accountability and human oversight in AI-assisted decisions.
- Analyze a real-world AI system using the principles of Trustworthy AI.

1. OVERVIEW

The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) algorithm is a risk assessment system developed by Northpointe Inc. (now Equivant). It is designed to estimate the likelihood that a defendant may commit another crime after release, a concept known as recidivism.

The system has been used by courts in several U.S. states to assist judges in decisions related to bail, sentencing, probation, and parole.

In 2016, ProPublica published an investigative report claiming that COMPAS produced racially disparate outcomes. The report generated international discussion about AI fairness, transparency, accountability, and ethics.

Today, COMPAS is considered one of the most important case studies in Trustworthy AI.

2. BACKGROUND

Judges must often decide whether a defendant should receive bail, probation, or a prison sentence. These decisions involve evaluating the likelihood that an individual may commit another crime in the future.

To support this process, AI-based risk assessment tools such as COMPAS were developed.

The COMPAS system evaluates multiple factors, including:

- Criminal history
- Age
- Previous offenses
- Behavioral information
- Social background
- Responses to questionnaires

Using these inputs, the system generates a risk score that helps judges evaluate future risk.

Importantly, COMPAS was intended to support judicial decision-making rather than replace judges.

3. TIMELINE

YEAR	EVENT
1998	Northpointe began developing the COMPAS system.
2000s	COMPAS was adopted by courts in several U.S. states.
2016	ProPublica published its investigation into COMPAS fairness.
2016	Northpointe (now Equivant) responded to the findings and defended the system.
2016–Present	COMPAS became a widely discussed case study in AI fairness and ethics.

4. PROBLEM STATEMENT

Criminal courts face several challenges:

- Making consistent judicial decisions.
- Assessing the likelihood of future criminal behavior.
- Reducing subjective judgment.
- Supporting public safety while protecting individual rights.

AI-based risk assessment tools were introduced to assist judges by providing structured risk evaluations.

The challenge was ensuring that these AI-generated predictions remained fair, transparent, and trustworthy.

5. HOW THE AI SYSTEM WORKED

Although the complete internal model is proprietary, the general process is publicly understood.

The AI system followed these steps:

- Collect information about the defendant.
- Analyze criminal history and other relevant factors.
- Generate a risk score (Low, Medium, or High).
- Provide the score to judges as decision-support information.
- Judges consider the score together with legal evidence before making a final decision.

The AI system was not intended to make judicial decisions independently.

6. WHAT WENT WRONG?

The controversy centered on fairness and transparency.

ProPublica's Findings

According to ProPublica's analysis:

- Some Black defendants were more frequently classified as high risk but did not reoffend (false positives).
- Some White defendants were more frequently classified as low risk but later reoffended (false negatives).

These findings suggested that prediction errors differed across demographic groups.

Developer's Response

Northpointe (now Equivant) disagreed with ProPublica's conclusions. The company argued that COMPAS satisfied predictive parity, meaning individuals with the same risk score had similar observed recidivism rates across groups.

This disagreement demonstrated that different definitions of fairness can produce different conclusions.

The COMPAS case remains an important example of how fairness in AI is a complex technical, legal, and ethical issue.

7. TRUSTWORTHY AI PRINCIPLES INVOLVED

Fairness

The case highlighted that AI systems should be evaluated for unfair outcomes across different demographic groups.
Lesson: Organizations should evaluate AI systems using multiple fairness metrics before deployment.

Transparency

The internal details of the COMPAS model were not publicly available.
Lesson: Greater transparency enables independent evaluation, auditing, and public trust.

Accountability

Although AI assisted decision-making, judges and institutions remained responsible for the final decisions.
Lesson: Organizations—not AI systems—are accountable for AI-assisted outcomes.

Human Oversight

Judges retained authority over all judicial decisions.
Lesson: AI should support human decision-making rather than replace it in high-impact domains.

Governance

The case demonstrated the importance of continuous monitoring, auditing, documentation, and responsible AI governance.
Lesson: Organizations should regularly evaluate AI systems for fairness, reliability, and compliance with legal and ethical standards.

8. HUMAN–AI INTERACTION PERSPECTIVE

COMPAS demonstrates why AI should function as a decision-support tool rather than an autonomous decision-maker.

Judges contribute legal expertise, contextual understanding, ethical reasoning, and professional judgment. AI contributes risk estimation, pattern recognition, and consistent analysis of large datasets.

Effective Human–AI Interaction requires:

- Human understanding of AI limitations.
- Appropriate trust in AI recommendations.
- Meaningful human oversight.
- Final decisions remaining under human control.

9. IMPACT

On the Criminal Justice System

- Increased discussion about fairness in AI.
- Greater interest in independent AI audits.
- More emphasis on transparency and accountability.

On AI Research

The COMPAS case became one of the most frequently studied examples of algorithmic fairness, responsible AI, AI ethics, explainable AI, and AI governance.

On Policymakers

Governments and researchers began promoting AI auditing, fairness testing, risk management, human oversight, and responsible AI frameworks.

10. LESSONS LEARNED

This case demonstrates that:

- Fairness is essential for AI systems used in criminal justice.
- Different fairness definitions may produce different conclusions.
- Transparency supports accountability and public trust.
- Human oversight remains necessary.
- Organizations must continuously evaluate AI systems.
- AI governance is essential for high-impact applications.

11. BEST PRACTICES

Organizations developing AI risk assessment systems should:

- Use representative datasets.
- Evaluate multiple fairness metrics.
- Conduct independent bias audits.
- Maintain meaningful human oversight.
- Improve transparency wherever possible.
- Monitor system performance continuously.
- Document model development and evaluation.
- Ensure compliance with legal and ethical requirements.

12. DISCUSSION QUESTIONS

- Why did the COMPAS system become controversial?
- Why is fairness difficult to measure in AI systems?
- Why is transparency important in criminal justice AI?
- Should AI systems influence judicial decisions?
- What role should human oversight play in AI-assisted justice?

13. KEY TAKEAWAYS

- AI fairness involves multiple technical definitions.
- Transparency improves trust and accountability.
- Human oversight is essential in high-impact AI applications.
- Organizations remain responsible for AI-assisted decisions.
- Trustworthy AI requires fairness, governance, accountability, and continuous monitoring.

14. CONNECTION TO PREVIOUS TOPICS

PREVIOUS TOPIC	CONNECTION
Trustworthy AI	Demonstrates fairness, accountability, transparency, and governance.
Human–AI Interaction	Shows AI as a decision-support tool rather than a replacement for judges.
Responsible AI	Highlights responsible deployment of AI in public services.
Explainable AI	Demonstrates why transparency improves public trust.

PREVIOUS TOPIC	CONNECTION
AI Governance	Shows the importance of auditing and continuous monitoring.

15. REFERENCES

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